

Yash Ravindra Charde

yashcharde.asu@gmail.com | +1 (602)394-3218 | linkedin.com/in/chardeyash/ | github.com/Yash-Charde

Summary

Software Engineer with 3+ years of experience building scalable web applications and backend services. Skilled in system design, analytical thinking, and delivering high-quality, production-ready solutions with measurable impact.

Education

Master of Science in Computer Science, Arizona State University, Tempe

Aug 2025 – May 2027

• **Coursework:** Cloud Computing, Data Processing at Scale, Data Mining, Statistical Machine Learning, Knowledge Representation, Data Visualization, Natural Language Processing

GPA: 4.00/4.00

Research Publications

• Co-authored paper on *Robust latent reasoning in large language models* under the supervision of Dr. Yanjie Fu. Under review at NeurIPS 2026 and the ICML 2026 (A. V. Malarkkan, M. R. Choudhury, U. Byahut, **Y. R. Charde**, V. Gupta, Y. Fu et al.)

Experience

Senior Member Technical, ADP Private Limited – Hyderabad, India

Apr 2024 – Jun 2025

- Developed a consolidated invoices page in RUN using React.js, Node.js, REST APIs, JavaScript, and Tailwind CSS, delivering a responsive and mobile-friendly UI with seamless cross-browser compatibility across multiple devices and platforms.
- Engineered an end-to-end Year-End Readiness Tracker using React.js and ASP.NET Web API, enabling clients to monitor W-2 and year-end progress, deadlines, and required actions through a centralized single-pane dashboard experience.
- Optimized year-end billing and production workflows, reducing billing processing time by 3 hours and completing W-2 production within 6 hours; migrated 48,000+ clients to Local Delivery, generating \$180K in annual cost savings.

Member Technical, ADP Private Limited – Hyderabad, India

Jul 2021 – Mar 2024

- Architected an event-driven integration platform for GSO to receive, persist, and asynchronously distribute all RUN business events to downstream security consumers, implementing reliable fan-out delivery to multiple subscriber types including REST endpoints, IBM MQ, and Kafka using consistent integration contracts for heterogeneous security tools.
- Maintained core processing components - MQReader (consumer & dispatcher) and BatchRetry (retry & reprocessing pipeline for transient failures) - ensuring at-least-once delivery and resilient event transfer across distributed messaging workflows.
- Implemented RED fraud alert email notifications for timely stakeholder awareness and escalation on security-sensitive events; led production support, incident triage, and root-cause analysis (RCA) to protect high-priority security event pipelines.
- Modernized the shipping details page by integrating delivery preferences with MVC architecture, streamlining workflows, reducing manual follow-ups, and delivering a more intuitive and responsive user experience.

Projects

Federated Learning System on AWS Cloud and Edge

- Architected an end-to-end Federated Learning pipeline on AWS using 10 distributed EC2 workers and an event-driven Lambda aggregator (triggered via S3) to collaboratively train a LeNet-5 CNN on MNIST, achieving 97.69% accuracy with consistent round-over-round convergence while eliminating polling overhead and preserving data locality.
- Extended the system to edge computing with AWS IoT Greengrass by deploying custom MQTT-triggered components across devices, scoped permissions for secure model uploads and sustaining 97.39% accuracy in a Greengrass-based deployment.

Scalable Cloud-Native Face Recognition Pipeline on AWS

- Designed a multi-tier elastic face recognition application on AWS with a stateless EC2 web tier, SQS-decoupled PyTorch inference workers, and S3 + DynamoDB for image storage and result persistence, sustaining sub-second average latency across 100 concurrent requests while cleanly scaling queues from 0 to peak load and back.
- Implemented a custom autoscaling controller in Python that dynamically scales the application tier from 0 to 15 EC2 instances based on SQS queue depth, with DynamoDB result caching delivering ~3x warm-cache speedup.

Technologies

Languages: C/C++, C#, Java, JavaScript (ES6+), TypeScript, Python, PL/SQL, SQL (MySQL, PostgreSQL), HTML, CSS

Frameworks & Libraries: React.js, Angular (v11-v12, TypeScript-based), Node.js, Express.js, ASP.NET Core, Microservices, RESTful APIs, MVC, Tailwind CSS, Cypress, D3.js, Pandas, NumPy, Matplotlib, OpenCV, Scikit-learn, TensorFlow, PyTorch

Tools & Platforms: AWS (EC2, S3, Lambda, DynamoDB, API Gateway), Docker, Kubernetes (basic), Jenkins, CI/CD pipelines, Git, GitHub, Bitbucket, Agile, Jira, Postman, Splunk, Figma, Linux/Ubuntu

Leadership & Recognition

- Recognized as Team of the Quarter for leading the Billing and Invoices initiative at ADP Pvt. Ltd.
- Awarded the “Win as One” recognition for contributions to the Year-End Health Tracker initiative at ADP Pvt. Ltd.